AI for Medical Imaging

Master Artificial Intelligence

Hoel Kervadec

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Universiteit van Amsterdam
Instituut voor Informatica

Amsterdam UMC

2026/09/10 – Segmentation (Part 2)

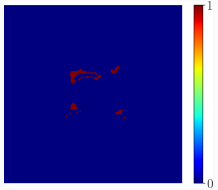
Today's plan

- CE loss derivation
- Network architectures, pre/post-processing, subsampling
- Break
- Weak supervision, other losses, other approaches
- Assigments and rubric on Canvas

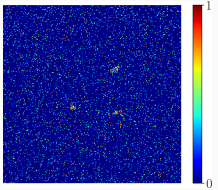
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Losses and gradient (cont.)

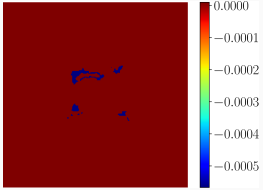
Gradients



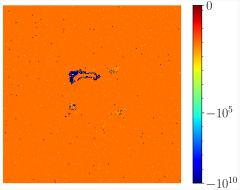
$y^{(1,k)}$



$s_\theta^{(1,k)}$



$\frac{\partial \mathcal{L}_{\text{DSC}}}{\partial s_\theta^{(0,k)}} + \frac{\partial \mathcal{L}_{\text{DSC}}}{\partial s_\theta^{(1,k)}}$



$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial s_\theta^{(0,k)}} + \frac{\partial \mathcal{L}_{\text{CE}}}{\partial s_\theta^{(1,k)}}$

$$\frac{\partial \mathcal{L}_{\text{DSC}}}{\partial s_\theta^{(i,k)}} = \begin{cases} \frac{-2\left(U^{(k)} - I^{(k)}\right)}{\left(U^{(k)}\right)^2} & \text{if } y^{(i,k)} = 1, \\ \frac{2I^{(k)}}{\left(U^{(k)}\right)^2} & \text{otherwise,} \end{cases}$$

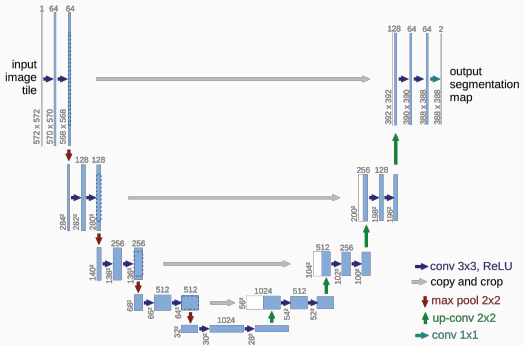
$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial s_\theta^{(i,k)}} = \begin{cases} \frac{-1}{s_\theta} & \text{if } y^{(i,k)} = 1, \\ 0 & \text{otherwise.} \end{cases}$$

with $I^{(k)} = \sum_{i \in \Omega} y^{(i,k)} s_\theta^{(i,k)}$ and $U^{(k)} = \sum_{i \in \Omega} \left[y^{(i,k)} + s_\theta^{(i,k)} \right]$.

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Architecture tuning and other levers

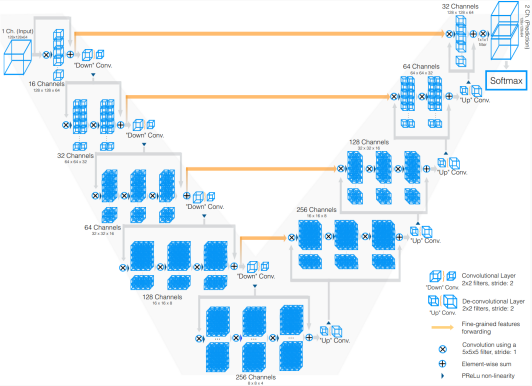
Architecture (i) – 2D



U-net: Convolutional networks for biomedical image segmentation

Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. Medical image computing and computer-assisted intervention —- MICCAI 2015.

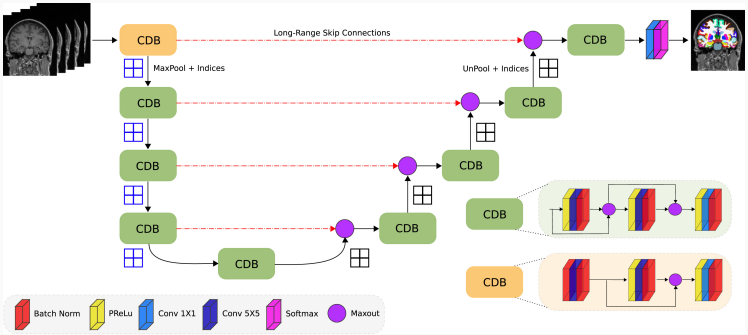
Architecture (ii) – 3D



V-net: Fully convolutional neural networks for volumetric medical image segmentation

Milletari, Fausto, Nassir Navab, and Seyed-Ahmad Ahmadi. 2016 fourth international conference on 3D vision (3DV). Ieee, 2016.

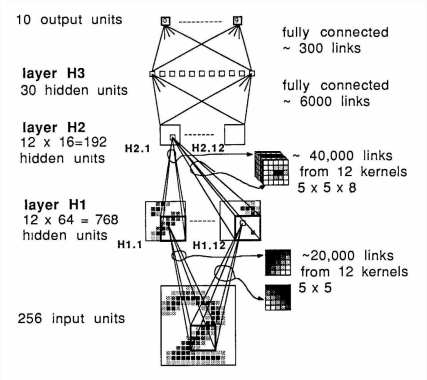
Architecture (iii) — 2.5D

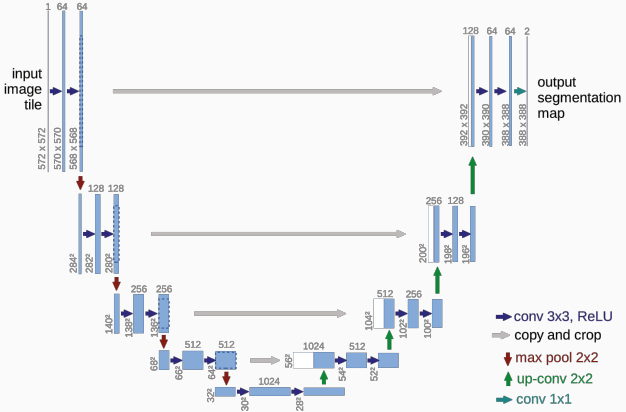


FastSurfer – A fast and accurate deep learning based neuroimaging pipeline

Henschel, Leonie, et al., NeuroImage 219 (2020): 117012.

Tuning the convolutions





Metadata and pre-processing

Can have diverse “spacings”/resolution within the same dataset

Might require or benefit from resampling (upsampling, downsampling)

Data augmentation

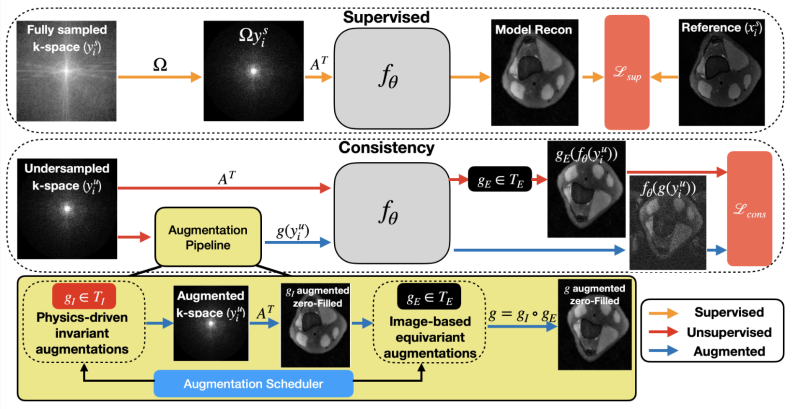
Random modifications of input images:

$$\mathcal{A}(x, y) = (x_a, y_a) \tag{3}$$

$$= \begin{cases} (\texttt{flip}(x, y)) & \text{if } r = \texttt{flip} \\ (\texttt{noise}(x), y) & \text{if } r = \texttt{noise} \\ (\texttt{rotation}(x, y)) & \text{if } r = \texttt{rotation} \\ ((x, y)) & \text{if } r = \texttt{id} \end{cases} \tag{4}$$

$$r \sim \mathcal{U}\{\texttt{id}, \texttt{flip}, \texttt{noise}, \texttt{rotation}, ...\} \tag{5}$$

“Physics” based augmentation



VORTEX: Physics-Driven Data Augmentations Using Consistency Training for Robust Accelerated MRI Reconstruction

Arjun D Desai, et al., MIDL 2022, https://2022.midl.io/papers/b_l_3

Offline vs online augmentation

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$$

"Offline":

$$\begin{aligned}\mathcal{D}_a &= \cup\{\mathcal{A}(\mathcal{D})\}^A \\ &= \cup\{\{A(x_n, y_n)\}_{n=1}^{N}\}^A\end{aligned}$$

$|\mathcal{D}| = N, |\mathcal{D}_a| = AN$: expand the dataset

"Online", at each iteration:

$$\begin{cases}(x_a, y_a) = \mathcal{A}(x_n, y_n) \\ s_\theta = \text{softmax}(\mathcal{N}(x_a; \theta)) \\ \mathcal{L}(y_a, s_\theta)\end{cases}$$

The network never sees twice the same image

Overview of training

Quick notation: $\min_\theta \quad \mathcal{L}(y, s_\theta)$

In practice: $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$

Each step:

$$\mathcal{B} \sim \mathcal{U}(\mathcal{D}) = \{(\check{x}_b, \check{y}_b)\}_{b=1}^B, \check{x}_b \subset x_b, \check{y}_b \subset y_b$$

$$\mathcal{L} = \sum_{b \in \mathcal{B}} \mathcal{L}(\check{y}_b, \check{s}_{\theta,b})$$

$$\mathcal{L}_{\text{CE}} = \sum_{b \in \mathcal{B}} \sum_{k \in \mathcal{K}} \sum_{i \in \check{\Omega}_b} -y_b^{(i,k)} \log(s_{\theta,b}^{(i,k)})$$

$$\theta_{t+1} = \texttt{optimizer}\left(\theta_t, \frac{\partial \mathcal{L}}{\partial \theta}\right)$$

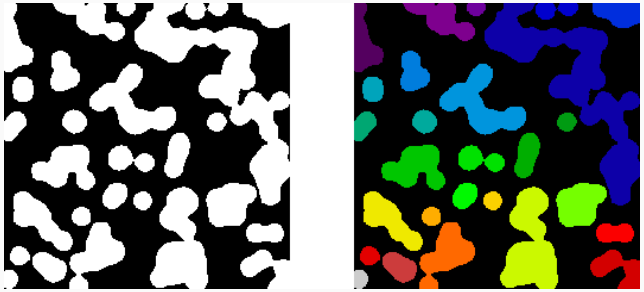
The network never sees the "whole picture".

Impact of sub-patching

Many patches can (will) end-up empty for some classes.

Watch-out for corner case of your losses.

Post-processing – Non-maxima suppression

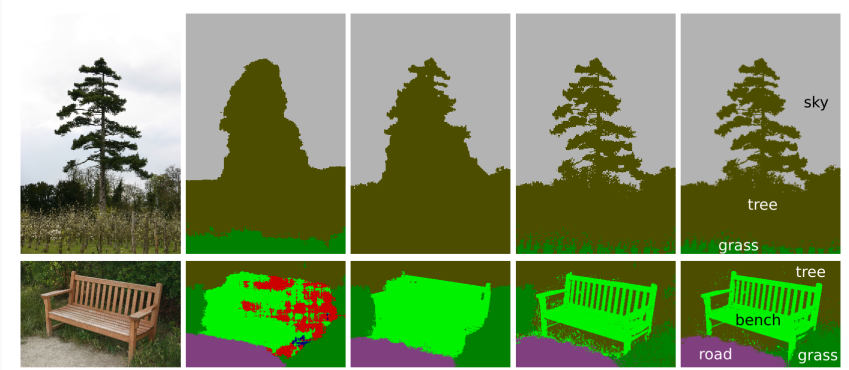


Can identify (here: color) connected components

`https://scikit-image.org/docs/stable/api/skimage.measure.html#skimage.measure.label`

Depending on the task: keep only the biggest one, set the rest to `background`.

Post-processing – DenseCRF (i)



(a) Image (b) Unary classifiers (c) Robust P^n CRF (d) Fully connected CRF, MCMC inference, 36 hrs (e) Fully connected CRF, our approach, 0.2 seconds

Efficient inference in fully connected crfs with gaussian edge potentials

Krähenbühl, Philipp, and Vladlen Koltun, Advances in neural information processing systems 24 (2011).

Post-processing – DenseCRF (ii)

$$\min_{S} \quad \sum_{i \in \Omega} \phi_{\text{unary}}(S^{(i)}; s_\theta) + \sum_{i,j \in \Omega^2} \phi_{\text{pairwise}}(S^{(i)}, S^{(j)}; X)$$

$$\phi_{\text{unary}}(S^{(i)}; s_\theta) := -\log\left(s_\theta^{(p, S^{(i)})}\right)$$

$$\phi_{\text{pairwise}}(S^{(i)}, S^{(j)}; X) := \mathbf{1}_{[S^{(i)} = S^{(j)}]} \sum_{m=1}^{2} w_m \kappa_m(i,j)$$

$$\kappa_1(i,j) := e^{-\frac{|\Omega_i - \Omega_j|^2}{2\omega_\alpha^2} - \frac{|x(i) - x(j)|^2}{2\omega_\beta^2}}$$

$$\kappa_2(i,j) := e^{-\frac{|\Omega_i - \Omega_j|^2}{2\omega_\gamma^2}}$$

Ω_i: pixel coordinates, $x(p)$: pixel intensity, $w_m \in \mathbb{R}$.

κ_1: appearance kernel for local consistency; κ_2: smoothness kernel, to remove noise.

Ensembling

Method to improve performances and reliability, popular for challenges

Train same network (or variants) several times, and average/combine their predictions

Downside: *very* costly

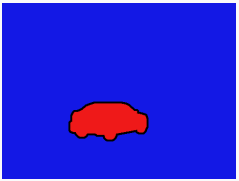
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Weak supervision

Weak labels



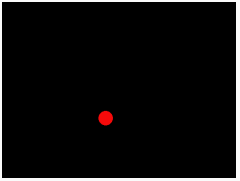
Image



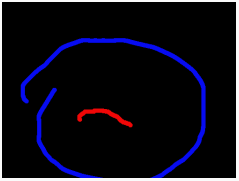
Full mask

There is a car

Tags



Dot



Scribbles

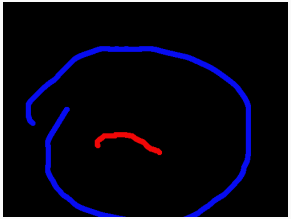


Bounding box

Proposal based methods vs direct losses



Input image x



Weak mask $\tilde{y}$

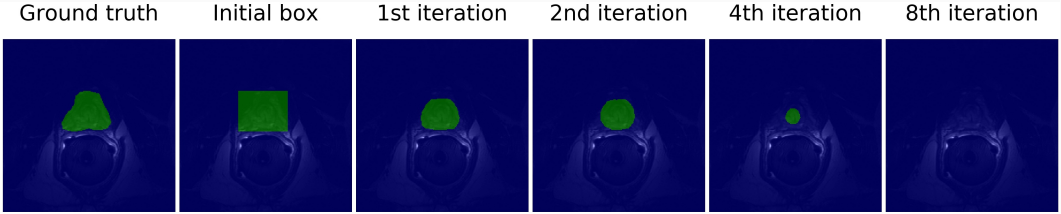
Proposal (y_θ):

$$\min_{\theta, y_\theta} \quad \mathcal{L}(y_\theta, s_\theta)$$

Direct loss $\mathcal{L}$, $\lambda \in \mathbb{R}$:

$$\min_{\theta} \quad \widetilde{\mathcal{L}}(\tilde{y}, s_\theta) + \lambda \mathcal{R}(s_\theta)$$

Issues with proposal based methods



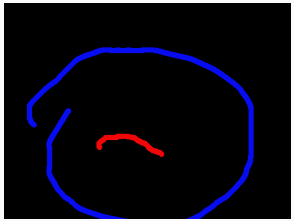
Deepcut: Object segmentation from bounding box annotations using convolutional neural networks

Rajchl, Martin, et al., IEEE transactions on medical imaging 36.2 (2016): 674-683.

Partial cross-entropy



Input image x



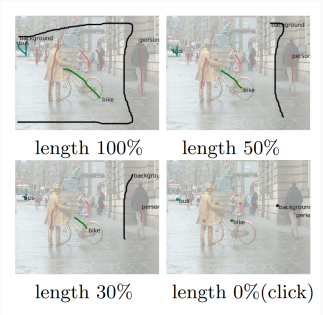
Weak mask $\tilde{y}$

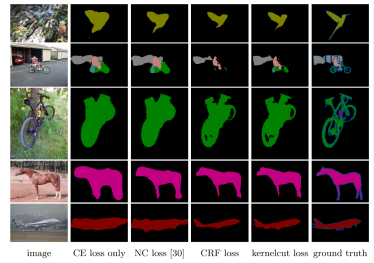
$\widetilde{\Omega} \subset \Omega$: subset of annotated pixels, $\tilde{y} \in \{0,1\}^{\Omega} \subset (\widetilde{\Delta}^K)^{\Omega}$, $\widetilde{\Delta}^K = \hat{\Delta}^K \cup \mathbf{0}$: "relaxed" simplex.

$$\widetilde{\mathcal{L}}_{\mathsf{CE}}(\tilde{y}, s_\theta) = \sum_{i \in \widetilde{\Omega}} \sum_{k \in \mathcal{K}} -y^{(i,k)} \log(s_\theta^{(i,k)})$$

$$= \sum_{i \in \Omega} \sum_{k \in \mathcal{K}} -\tilde{y}^{(i,k)} \log(s_\theta^{(i,k)})$$

CRF loss



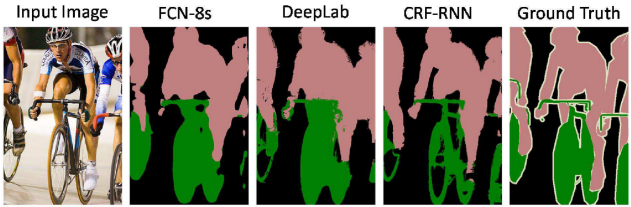


$$\min_{\theta} \quad \widetilde{\mathcal{L}}_{\mathsf{CE}}(\tilde{y}, s_\theta) + \lambda \mathsf{CRF}(s_\theta; x)$$

On regularized losses for weakly-supervised cnn segmentation

Tang, Meng, et al., Proceedings of the European conference on computer vision (ECCV). 2018.

CRF loss (cont.)



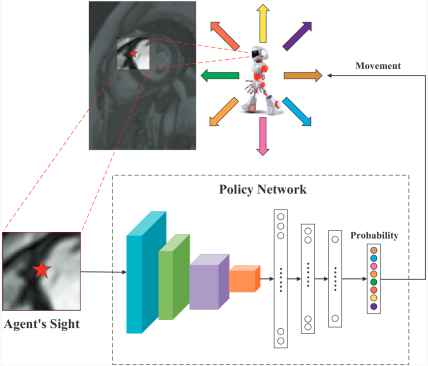
Conditional random fields as recurrent neural networks

Zheng, Shuai, et al., CVPR 2015.

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Extra – other various approaches

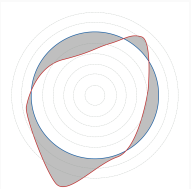
Robot's pathfinding



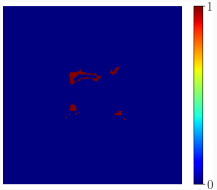
Left Ventricle Contouring in Cardiac Images Based on Deep Reinforcement Learning

Sixing Yin, Yameng Han, Judong Pan, Yining Wang, Shufang Li, MIDL 2022,
https://2022.midl.io/papers/a1

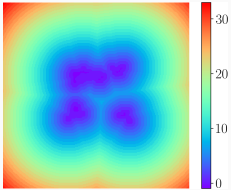
Boundary loss



$\partial\Omega_y$, $\partial\Omega_s$



$y^{(1,k)}$



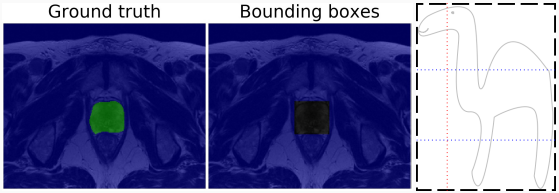
$\frac{\partial \mathcal{L}_B}{\partial s_\theta^{(1,k)}} = \phi^{(i,k)}_{\partial\Omega_y^{(k)}}$

Minimize the distance between the label boundary ($\partial\Omega_y$) and the predicted boundary ($\partial\Omega_s$) through the signed distance map ϕ: $\mathcal{L}_B(y, s_\theta) = \sum_{k\in\mathcal{K}} s_\theta^{(i,k)} \phi^{(i,k)}_{\partial\Omega_y^{(k)}}$

Boundary loss for highly unbalanced segmentation

H Kervadec, J Bouchtiba, C Desrosiers, É Granger, J Dolz, I Ben Ayed, 2020, Medical Image Analysis.

Tightness prior for bounding boxes

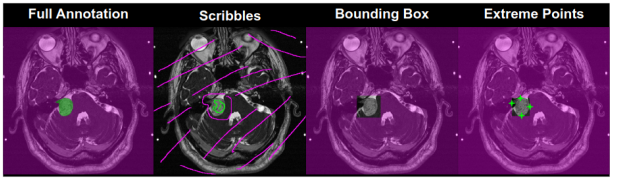


Exploit the prior knowledge of a box, notably its tightness, to design a series of losses to balance the lack of accurate annotations.

Bounding boxes for weakly supervised segmentation: Global constraints get close to full supervision

H Kervadec, J Dolz, S Wang, E Granger, I Ben Ayed , 2020, Medical Imaging with Deep Learning (MIDL).

Extreme points with geodesic distances



Inter extreme points geodesics for end-to-end weakly supervised image segmentation

Dorent, R., Joutard, S., Shapey, J., Kujawa, A., Modat, M., Ourselin, S., & Vercauteren, T. (2021, September). In International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 615-624). Cham: Springer International Publishing.).

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Notation (updated)

Summary of notation (so far) i

$\Omega \subset \mathbb{R}^D$: image space of dimension D

$x \in \mathbb{R}^{M \times \Omega}$: input image with M modalities

$\mathcal{K} = \{0 : \texttt{background}, 1 : \texttt{class-1}, ..., K : \texttt{class-K}\}$: set of K+1 classes to predict

$Y \in \mathcal{K}^\Omega$: ground-truth, $S \in \mathcal{K}^\Omega$ prediction

θ: network parameters to "learn"

$\mathcal{N}(x; \theta) \in \mathbb{R}^{(K+1) \times \Omega}$: raw output of the neural network

Δ^K: K-simplex (so $K + 1$ probabilities drawn in a K-dimensional space)

$\widehat{\Delta}^K$: K-simplex, restricted to its vertices (one-hot).

Summary of notation (so far) ii

$Y \in \mathcal{K}^\Omega \underset{\text{one-hot}}{\longmapsto} y \in (\widehat{\Delta}^K)^\Omega \subset \{0,1\}^{(K+1)\times\Omega}$: one-hot encoding of the label Y

$\mathcal{N}(x;\theta) \underset{\text{softmax}}{\longmapsto} s_\theta \in (\Delta^K)^\Omega \subset [0,1]^{(K+1)\times\Omega}$: vector of predicted probabilities after softmax layer

$s_\theta \in (\Delta^K)^\Omega \underset{\text{argmax}}{\longmapsto} S \in \mathcal{K}^\Omega \underset{\text{one-hot}}{\longmapsto} s \in (\widehat{\Delta}^K)^\Omega$: representation of the prediction as one-hot vector (useful for indicator functions)

$\mathcal{L}(s_\theta, y)$: some loss

$\mathcal{L}_{\text{CE}}$: cross-entropy loss

$\mathcal{L}_{\text{DSC}}$: Dice loss

$\nabla\theta = \frac{\partial \mathcal{L}(s_\theta, y)}{\partial \theta}$: gradient

Summary of notation (so far) iii

γ: learning rate

$\mathcal{A}(x, y) = (x_a, y_a)$: data augmentation function (non-deterministic)

$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$: dataset with all annotated samples

$\mathcal{D}_a = \cup\{\{A(x_n, y_n)\}_{n=1}^{N}\}^{A}$: dataset augmented "offline"

$\mathcal{B} \sim \mathcal{U}(\mathcal{D}) = \{(\breve{x}_b, \breve{y}_b)\}_{b=1}^{B}, \breve{x}_b \subset x_b, \breve{y}_b \subset y_b$: batch (subset of dataset) of sub-patches of samples

$\breve{\Omega}_b \subset \Omega_b$: image space subset of the sub-patched sample (2D slice, 3D patch)

$\mathcal{R}(s_\theta)$: (unsupervised) regularizer

$\mathrm{CRF}(s_\theta; x)$: CRF conditioned on the input image x

Summary of notation (so far) iv

$\widetilde{\mathcal{L}}(\tilde{y}, s_\theta)$: a loss designed for a weak label